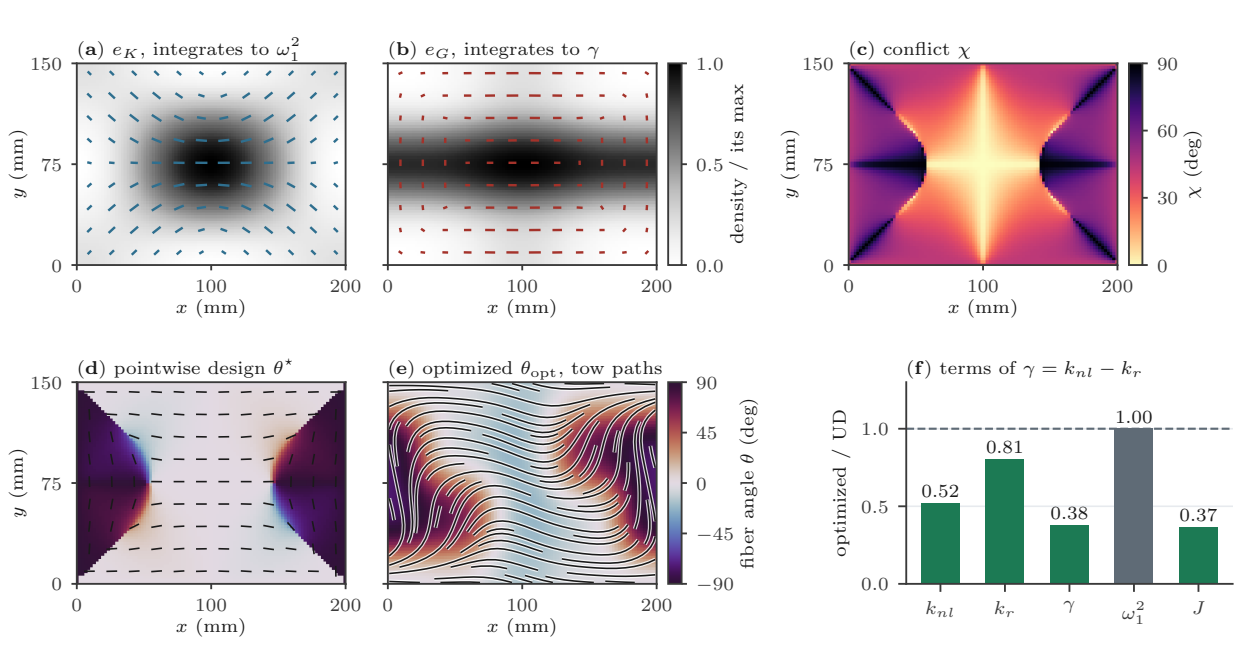

(a) e_K, integrates to ω_1^2
(b) e_G, integrates to γ
(c) conflict χ
(d) pointwise design $\theta^\star$
(e) optimized θ_{opt}, tow paths
(f) terms of $\gamma = k_{nl} - k_r$
y (mm)
x (mm)
density / its max
χ (deg)
fiber angle θ (deg)
optimized / UD
0.52
0.81
0.38
1.00
0.37
k_{nl}
k_r
γ
ω_1^2
J